

Brain Tumor MRI Classification and Prediction

Deep Learning Based Brain Tumor Detection Using MRI Scans

A Professional Research Report Submitted By

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Certificate

This is to certify that the project report entitled "**Brain Tumor MRI Classification Prediction: Deep Learning Based Brain Tumor Detection Using MRI Scans**" submitted by **Basim Hassan** is a bonafide work carried out under our supervision during the academic year 2024–2026.

The work presented in this report is original and fulfills the academic requirements for the completion of the Class 12 Bio Science research project.

Guide 1: Abdul Muhammed Rasheed

Guide 2: Amal Ajayan

Declaration

I hereby declare that this report entitled "**Brain Tumor MRI Classification and Prediction: Deep Learning Based Brain Tumor Detection Using MRI Scans**" is an original work carried out by me under the guidance of my mentors.

The work has not been submitted elsewhere for any degree, diploma, or academic certification.

Basim Hassan

Acknowledgement

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Abstract

Brain tumor diagnosis using MRI imaging is a challenging and critical problem in medical science. Manual diagnosis requires expert radiologists and considerable time, making automated systems highly valuable in modern healthcare.

This project presents a deep learning-based brain tumor classification system capable of categorizing MRI images into four classes: Glioma, Meningioma, Pituitary Tumor, and No Tumor. The proposed system utilizes Convolutional Neural Networks (CNNs) and Transfer Learning with MobileNetV2 to achieve high classification accuracy.

The project was implemented using TensorFlow and Keras in Google Colab. Advanced preprocessing techniques such as image augmentation, normalization, dropout regularization, batch normalization, and L2 regularization were applied to improve performance and reduce overfitting.

The final refined model achieved approximately 88.19% test accuracy with a test loss of 0.5269, demonstrating the effectiveness of transfer learning in medical image classification.

This work highlights the potential of artificial intelligence in supporting healthcare diagnostics and improving early detection of brain tumors.

Keywords: Brain Tumor Classification, MRI, Deep Learning, CNN, Transfer Learning, MobileNetV2, TensorFlow, Keras, Medical Imaging, Artificial Intelligence

Chapter 1

Introduction

Brain tumors are abnormal growths of cells inside the brain that can severely affect neurological functioning and human health. Early diagnosis and treatment significantly improve survival rates and patient outcomes.

Magnetic Resonance Imaging (MRI) is one of the most effective imaging techniques used for detecting brain abnormalities due to its high-resolution visualization of soft tissues.

Traditional diagnosis methods rely heavily on experienced radiologists. However, manual analysis of MRI images is time-consuming and susceptible to human error. Artificial Intelligence and Deep Learning provide powerful alternatives for automated image analysis and tumor classification.

Deep learning models, especially Convolutional Neural Networks (CNNs), have shown remarkable performance in image recognition tasks. Transfer learning models such as MobileNetV2 can leverage pre-trained knowledge from large-scale datasets to improve medical imaging performance.

The primary objective of this project is to develop a robust deep learning-based MRI brain tumor classification system capable of accurately identifying different tumor categories.

1.1 Objectives

- Develop an AI-based brain tumor classification system.
- Classify MRI images into four tumor categories.
- Reduce overfitting using regularization techniques.
- Improve model accuracy using transfer learning.
- Analyze training and validation performance.

Chapter 2

Literature Survey / Related Work

Recent advancements in deep learning have significantly improved medical image classification systems.

Several studies have demonstrated the effectiveness of Convolutional Neural Networks in detecting abnormalities from MRI scans. Traditional CNN architectures achieved promising results but often required large datasets and high computational power.

Transfer learning emerged as an effective technique to address limited medical datasets. Models such as VGG16, ResNet, EfficientNet, and MobileNetV2 have been widely adopted.

MobileNetV2 is particularly effective because of its lightweight architecture, reduced computational complexity, and strong feature extraction capabilities.

Researchers have also explored data augmentation, dropout regularization, and ensemble methods to improve model robustness and generalization.

Despite significant progress, challenges remain in handling dataset imbalance, varying MRI quality, and generalization across hospitals.

This project contributes by combining transfer learning, advanced regularization, and augmentation techniques into a comprehensive MRI tumor classification pipeline.

Chapter 3

Methodology

3.1 Data Collection

The MRI dataset used in this project consisted of approximately 5600 training images and 1600 testing images categorized into four classes:

- Glioma
- Meningioma
- Pituitary
- No Tumor

The dataset was organized into separate training and testing directories.

3.2 Data Preprocessing

The following preprocessing techniques were applied:

- Image resizing to 224×224 pixels
- Pixel normalization
- Rescaling pixel values to $[0,1]$

3.3 Data Augmentation

To increase dataset diversity and reduce overfitting, augmentation techniques were used:

- Rotation
- Width shifting

- Height shifting
- Shearing
- Zooming
- Horizontal flipping

3.4 Model Architecture

Initially, a custom CNN architecture was developed. Later, MobileNetV2 transfer learning was implemented for better feature extraction and improved accuracy.

3.5 Transfer Learning with MobileNetV2

MobileNetV2 pre-trained on ImageNet was used as the base model.

The top classification layers were removed and replaced with:

- Global Average Pooling Layer
- Dense Layer
- Dropout Layer
- Softmax Output Layer

3.6 Training Strategy

The model was trained using:

- Optimizer: Adam
- Loss Function: Categorical Crossentropy
- Epochs: 30
- Batch Size: 32

Callbacks used:

- EarlyStopping
- ReduceLROnPlateau

3.7 Regularization Techniques

To improve generalization and reduce overfitting:

- Dropout Regularization
- Batch Normalization
- L2 Regularization

Chapter 4

Results and Discussion

4.1 Performance Metrics

Metric	Value
Test Accuracy	88.19%
Test Loss	0.5269
Epochs	30
Framework	TensorFlow/Keras
Base Model	MobileNetV2

Table 4.1: Final Model Performance

4.2 Training Analysis

The training accuracy consistently improved over epochs, indicating successful learning from the MRI dataset.

Validation accuracy also improved significantly after applying transfer learning and regularization techniques.

The gap between training and validation metrics was reduced after implementing dropout and batch normalization.

4.3 Training Graphs

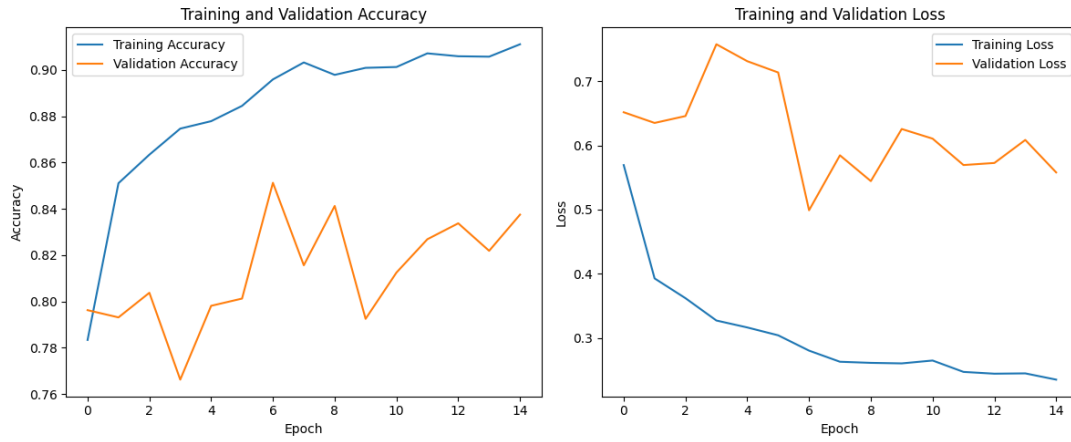


Figure 4.1: Training and Validation Accuracy/Loss

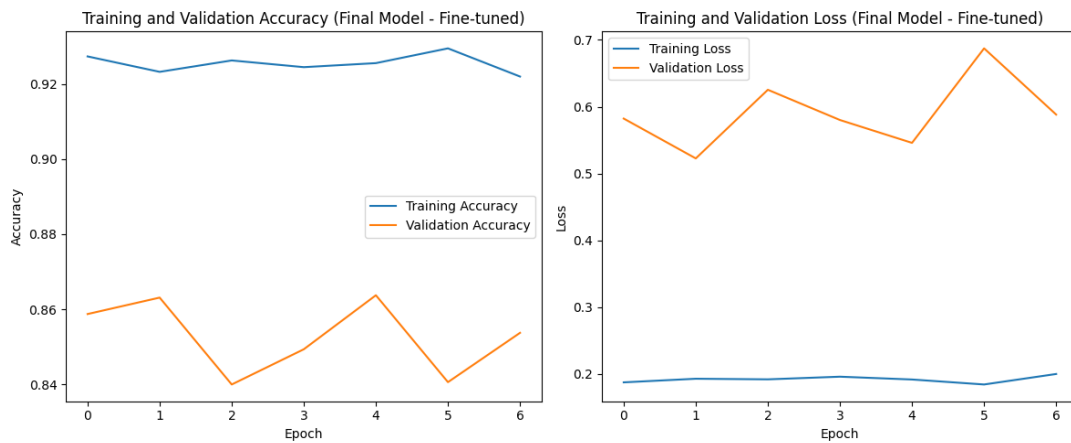


Figure 4.2: Fine Tuned Model Performance

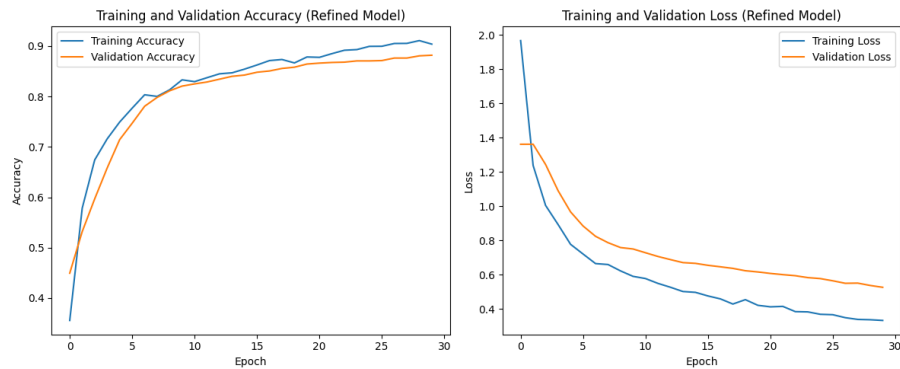


Figure 4.3: Refined Model Accuracy

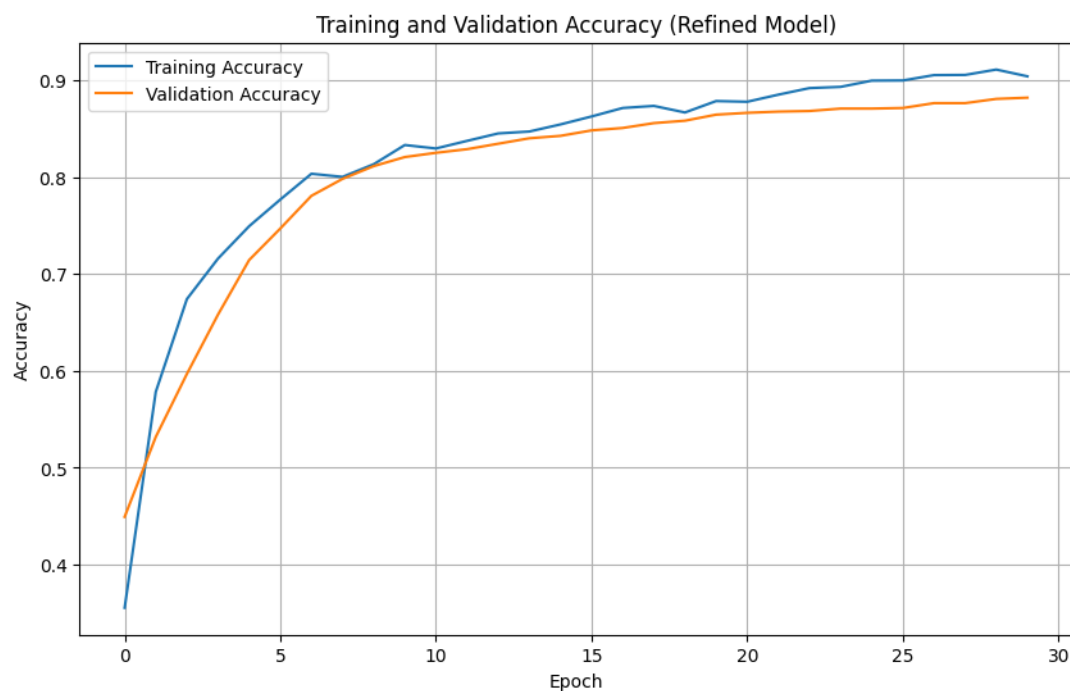


Figure 4.4: Additional Training Metrics

4.4 Discussion

The MobileNetV2-based architecture significantly outperformed the custom CNN model.

Transfer learning improved feature extraction capabilities while reducing training time.

Data augmentation improved generalization by increasing image diversity.

The refined model achieved strong classification performance suitable for AI-assisted healthcare research.

Chapter 5

Conclusion

This project successfully developed a deep learning-based MRI brain tumor classification system using TensorFlow and MobileNetV2 transfer learning.

The model demonstrated strong performance in classifying MRI images into Glioma, Meningioma, Pituitary, and No Tumor categories.

Advanced regularization and augmentation techniques improved model generalization and reduced overfitting.

The final refined model achieved approximately 88.19% accuracy, highlighting the effectiveness of transfer learning in medical image analysis.

This project demonstrates the growing importance of Artificial Intelligence in healthcare diagnostics and medical imaging.

Chapter 6

Future Scope

Future improvements may include:

- EfficientNet and ResNet integration
- Hyperparameter optimization
- Ensemble learning methods
- 3D CNN architectures for NIfTI MRI data
- Larger and more diverse datasets
- Real-time web deployment
- AI-assisted clinical support systems

Chapter 7

References

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Appendix A

Appendix: Source Code

A.1 Google Drive Mounting

```
1 from google.colab import drive
2 drive.mount('/content/drive')
```

A.2 Data Preprocessing and Augmentation

```
1 from tensorflow.keras.preprocessing.image import
   ImageDataGenerator
2
3 train_datagen = ImageDataGenerator(
4     rescale=1./255,
5     rotation_range=20,
6     width_shift_range=0.2,
7     height_shift_range=0.2,
8     shear_range=0.2,
9     zoom_range=0.2,
10    horizontal_flip=True,
11    fill_mode='nearest'
12 )
```

A.3 MobileNetV2 Transfer Learning

```
1 from tensorflow.keras.applications import MobileNetV2
2
3 base_model = MobileNetV2(
4     input_shape=(224,224,3),
```

```
5     include_top=False,
6     weights='imagenet'
7 )
8
9 base_model.trainable = False
```

A.4 Model Training

```
1 history = model.fit(
2     train_generator,
3     epochs=15,
4     validation_data=test_generator
5 )
```

A.5 Model Evaluation

```
1 loss, accuracy = model.evaluate(test_generator)
2 print(f"Test Accuracy: {accuracy:.4f}")
```

Appendix B

Appendix: Model Outputs

B.1 Training Logs

Epoch 1/15
accuracy: 0.7834
val_accuracy: 0.7962

Epoch 15/15
accuracy: 0.9111
val_accuracy: 0.8375

B.2 Model Summary

Total Parameters: 2,915,908
Trainable Parameters: 657,924
Non-trainable Parameters: 2,257,984

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Thank You

A handwritten signature in black ink, appearing to be 'Basim Hassan', enclosed within a circular flourish.